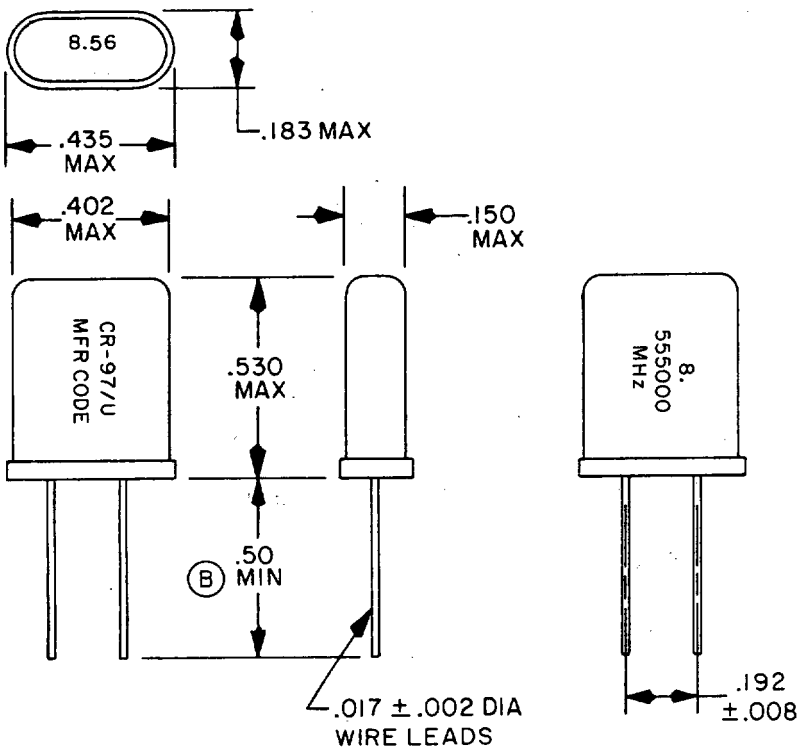
MILITARY SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR-97/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the crystal unit described herein shall consist of this specification and the latest issue of MIL-C-3098.

Pertinent characteristics: 8 to 10 MHz; fundamental; noncontrolled; antiresonance.



INCHES	MM
.002	.05
.008	.20
.017	.43
.150	3.81
.183	4.65
.192	4.88
.402	10.21
.435	11.05
.50	12.7
.530	13.46

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only and are based upon 1 inch = 25.4 mm.
3. The type designator, frequency, and manufacturer's code shall be not less than .038 (.97 mm) in height and not less than .027 (.69 mm) in width.

FIGURE 1. Crystal unit - CR-97/U.

Ⓑ denotes changes

FSC 5955

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Ⓑ TABLE I. Requirements.

Requirement	Specified value or condition
Holder	HC-18/U or HC-43/U (MIL-H-10056/9)
Dimensions, marking, and configuration	See figure 1.
Frequency range	8 to 10 MHz, inclusive
Capacitance, shunt	7 pF, maximum
Mode of oscillation	Fundamental
Government reference standard test set	TS-330/TSM
Rated drive level	10.0 ±2.0 mW
Calibration values:	
Resistance	11 ohms
Crystal current	30 mA
Antiresonance, load capacitance	32.0 ±0.5 pF
Operating temperature range (noncontrolled)	-40°C to ±85°C, inclusive
Frequency tolerance	±50 ppm
Equivalent resistance	35 ohms, maximum
Shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Vibration: 1/	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Thermal shock:	
Frequency change permitted	±5 ppm
Equivalent resistance change permitted	±10 percent
Aging:	
Frequency change permitted	±5 ppm

1/ Marking information: Acquisition documents shall specify the following.

a. The crystal frequency shall be stamped in megahertz (MHz) on side of the can in accordance with MIL-C-3098.

b. The crystal frequency shall also be stamped on top of the can in kilohertz (kHz).

2/ Method 204 of MIL-STD-202, test condition A.

NOTE: These crystal units are for use in the Collins 17L-7A Communication Transmitter.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 85
DLA - ES

User activities:
Air Force - 17, 19, 84

Agent:
DLA - ES

Preparing activity:
Army - ER

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